SIMATS ENGINEERING — Engineer to Excel

SIMATS — Saveetha Institute of Medical And Technical Sciences (Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-3

3. Water Jug Problem

Aim

To measure 2 liters of water using 4-liter and 3-liter jugs.

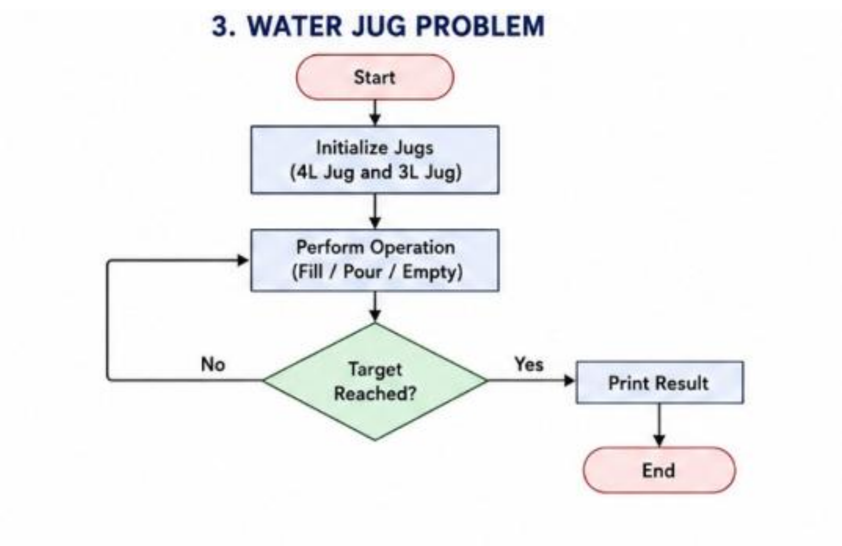
Objective

To obtain the required quantity of water by performing fill, pour, and empty operations.

Algorithm

1. Fill the 4-liter jug.
2. Pour water into the 3-liter jug.
3. Empty the 3-liter jug when full.
4. Repeat the process.
5. Stop when 2 liters remain in the 4-liter jug.

Flowchart



Code

```
jug1 = 4
jug2 = 3
target = 2
x = 0
y = 0
while x != target and y != target:
    if x == 0:
        x = jug1
    elif y == jug2:
        y = 0
    else:
        transfer = min(x, jug2 - y)
        x -= transfer
        y += transfer
    print("Jug1 =", x, "Jug2 =", y)

print("Target achieved!")
```

Output

```
File  Edit  Format  Run  Options  Window  Help
jug1 = 4
jug2 = 3
target = 2

a = 0
b = 0

while a != target and b != target:

    if a == 0:
        a = jug1          # Fill Jug 1
    elif b == jug2:
        b = 0             # Empty Jug 2
    else:
        t = min(a, jug2 - b)
        a -= t
        b += t            # Pour water

    print(a, b)

print("Target Achieved!")
```

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:1
on win32
Enter "help" below or click "Help" above for more info
>>>
==================================================== 
cal/Programs/Python/Python314/water jug.py ===========
========
4 0
1 3
1 0
0 1
4 1
2 3
Target Achieved!
>>>
```

Result

The program successfully measures 2 liters of water.

Conclusion

The Water Jug problem is solved using state-space operations.